Nonabelian Symmetries — Srednicki Ch. 24

Source: M. Srednicki, Quantum Field Theory, Chapter 24 (Nonabelian Symmetries), Problems 24.1–24.4.

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Antisymmetry of the Generator, Q24.1

Show that θ_{ij} in eq. (24.4) must be antisymmetric if R is orthogonal.

Solution. Consider an infinitesimal transformation of $SO(N)$,

$$R_{ij} = \delta_{ij} + \theta_{ij}. \tag{1.1}$$

By property of $SO(N)$, we have

$$R_{ij}R_{ik} = \delta_{jk} \implies (\delta_{ij} + \theta_{ij})(\delta_{ik} + \theta_{ik}) = \delta_{jk} + \theta_{jk} + \theta_{kj} = \delta_{jk} \implies \theta_{jk} = -\theta_{kj}. \tag{1.2}$$

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Structure Constants from SO(N) Transformations, Q24.2

By considering the SO(N) transformation $R'^{-1}R^{-1}R'R$, where R and R' are independent infinitesimal SO(N) transformations, prove eq. (24.7).

Solution. Now, we write the $\theta_{ij} = -i\theta^a(T^a)_{ij}$, where $a = 1, \ldots, N(N-1)/2$ and T^a is a set of basis Hermitian matrices with normalization condition

$$\mathrm{Tr}(T^aT^b) = 2\delta^{ab}. \tag{2.1}$$

Then, we consider

$$\begin{aligned} R'^{-1}R^{-1}R'R &= (1 + i\theta'^aT^a)(1 + i\theta^bT^b)(1 - i\theta'^cT^c)(1 - i\theta^dT^d) \\ &= 1 - 2\theta'^a\theta^bT^aT^b + 2\theta'^a\theta^bT^bT^a + \mathcal{O}(\theta^2). \end{aligned} \tag{2.2}$$

By the definition of group, the left-hand side is also an infinitesimal transformation

$$1 - i\theta''^c T^c = 1 - 2\theta'^a\theta^b T^a T^b + 2\theta'^a\theta^b T^b T^a + \mathcal{O}(\theta^2), \tag{2.3}$$

which gives

$$[T^a, T^b] = i\frac{\theta''^c}{\theta'^a\theta^b}T^c = if^{abc}T^c. \tag{2.4}$$

Noether Current and Charge Algebra, Q24.3

- Find the Noether current $j^{a\mu}$ for the transformation of eq. (24.6).
- Show that $[\phi_i, Q^a] = (T^a)_{ij}\phi_j$, where Q^a is the Noether charge.
- Use this result, eq. (24.7), and the Jacobi identity (see problem 2.8) to show that $[Q^a, Q^b] = if^{abc}Q^c$.

Solution. For scalar field theories that is invariant under $SO(N)$ ($\delta\mathcal{L} = 0$). The infinitesimal transformation is given by

$$\phi_i(x) \to \phi_i'(x) = \phi_i(x) - i\theta^a (T^a)_{ij}\phi_j(x). \tag{3.1}$$

Therefore, the Noether current is given by

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i(x))}\delta\phi_i(x) = i\theta^a\partial^\mu\phi_i(x)(T^a)_{ij}\phi_j(x). \tag{3.2}$$

Getting rid of the parameter, we find

$$j^{a\mu} = i\partial^\mu\phi_i(x)(T^a)_{ij}\phi_j(x). \tag{3.3}$$

Then, we can defined the conserved charge

$$Q^a = \int_\Sigma d^{d-1}x\, j^{a0} = -i(T^a)_{ij}\int_\Sigma d^{d-1}x\,\pi_i(x)\phi_j(x). \tag{3.4}$$

Using the equal time commutation relation,

$$[\phi_i(\mathbf{x},t), \pi_j(\mathbf{y},t)] = i\delta_{ij}\delta^{d-1}(\mathbf{x}-\mathbf{y}), \quad [\phi_i(\mathbf{x},t), \phi_j(\mathbf{y},t)] = 0, \tag{3.5}$$

we find

$$[\phi_i(\mathbf{x},t), Q^a] = (T^a)_{ij}\phi_j(\mathbf{x},t), \tag{3.6}$$

where we have used the fact that Q_a is conserved, and therefore the same for any t. Similarly

$$[\pi_i(\mathbf{x},t), Q^a] = -(T^a)_{ij}\pi_i(\mathbf{x},t). \tag{3.7}$$

Then, we consider

$$
\begin{aligned}
&= i(T^a)_{ij} \int_\Sigma d^{d-1}x \, [\pi_i(x)\phi_j(x), Q^b] \\
&= i(T^a)_{ij} \int_\Sigma d^{d-1}x \, \big(\pi_i(x)[\phi_j(x), Q^b] + [\pi_i(x), Q^b]\phi_j(x)\big) \\
&= i(T^a)_{ij} \int_\Sigma d^{d-1}x \, (\pi_i(x)(T^b)_{jk}\phi_k(x) - \pi_k(T^b)_{ki}\phi_j(x)) \\
&= i[T^a, T^b]_{ik} \int_\Sigma d^{d-1}x \, \pi_i(x)\phi_k(x) = if^{abc}Q^c.
\end{aligned}
\tag{3.8}
$$

Generators of Sp($2N$), Q24.4

The elements of the group SO(N) can be defined as $N \times N$ matrices R that satisfy

$$R_{ii'}R_{jj'}\delta_{i'j'} = \delta_{ij}. \tag{4.1}$$

The elements of the *symplectic group* Sp($2N$) can be defined as $2N \times 2N$ matrices S that satisfy

$$S_{ii'}S_{jj'}\eta_{i'j'} = \eta_{ij}, \tag{4.2}$$

where the *symplectic metric* η_{ij} is antisymmetric, $\eta_{ij} = -\eta_{ji}$, and squares to minus the identity: $\eta^2 = -I$. One way to write η is

$$\eta = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \tag{4.3}$$

where I is the $N \times N$ identity matrix. Find the number of generators of Sp($2N$).

Solution. We shall write down an infinitesimal transformation of $Sp(2N)$,

$$S_{ij} = \delta_{ij} - i\theta^a(T^a)_{ij}, \tag{4.4}$$

so that

$$(\delta_{ij} - i\theta^a(T^a)_{ij})(\delta_{lk} - i\theta^a(T^a)_{lk})\eta_{jk} = \eta_{il} + i\theta^a(T^a)_{lk}\eta_{ik} + i\theta^a(T^a)_{ij}\eta_{jl} = \eta_{il}. \tag{4.5}$$

This implies that

$$(T^a)_{lj}\eta_{ij} + (T^a)_{ij}\eta_{jl} = 0, \tag{4.6}$$

which means that $T^a\eta$ must be symmetric. Therefore, there are $N(2N+1)$ independent components for $T^a\eta$, and hence for T^a.